

SPECTRAL RESOLUTION OF GROUPS OF UNITARY OPERATORS

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Introduction. This paper discusses a spectral resolution theorem for groups of unitary operators, where the group involved is any locally compact Abelian topological group. If the group is specialized to be the real line our theorem becomes the well-known Stone theorem on 1-parameter groups of unitary operators, while if the group is specialized to be the integers it becomes the usual spectral resolution of a single unitary operator and its powers. We assume the Pontrjagin duality theorem, a nominal amount of operator theory (in Hilbert space) and the Herglotz-Bochner-Weil theorem on the representation of positive definite functions.

1. Preliminaries. We begin by establishing some notation which will be fixed throughout the first part of this paper. G will be a locally compact Abelian topological group (not necessarily separable) and G^* will be its character group. We shall use x and y for points in G and ξ and ζ for points in G^* ; for (complex-valued) functions defined on G we shall use f and g while for (complex-valued) functions defined on G^* we shall use φ and ψ . L_p and L_p^* will be the usual Banach spaces of (complex-valued) functions of integrable p -th power defined on G and G^* respectively while L_∞ and L_∞^* will be the Banach spaces of continuous functions (on G and G^* respectively) which, for each $\epsilon > 0$, are less than ϵ outside some compact set. By a Borel set we shall mean a set in the Borel field determined by the collection of open sets and by a Borel function a function which is measurable with respect to the Borel sets. The symbol $[x, \xi]$ will stand for the "character function" between G and G^* , i.e., that function on $G \times G^*$ which sets up the duality relation between G and G^* .

By a regular measure on G^* we shall mean a countably additive non-negative set function m defined for all Borel sets and having the following property: if B^* is any Borel set, then $m(B^*) = \inf m(O^*)$, this inf being taken over all open O^* which contain B^* . If $m(G^*) < \infty$ we say m is a finite measure. A function $f(x)$ on G is called positive definite if it is continuous and if for every finite set $x_1, \dots, x_n$ of points in G the matrix (a_{ij}) defined by $a_{ij} = f(x_i - x_j)$ is positive definite. In these terms we can state the Herglotz-Bochner-Weil representation theorem as follows:

I. $f(x)$ is positive definite if and only if there exists a finite regular measure m on G^* for which

$$f(x) = \int [x, \xi] m(d\xi).$$

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This statement is not quite the same as that given in the book of A. Weil [2] but can be derived easily from it.

By a complex measure we shall mean a set function m defined for all Borel sets, and of the form $m = m_1 - m_2 + i(m_3 - m_4)$, where the m_i are finite regular measures. We shall have occasion to use the following easily proved fact:

II. If $g(x)$ is any function of the form

$$g(x) = \int [x, \xi] m(d\xi),$$

where m is a complex measure, then the complex measure m is uniquely determined.

The group of operators whose spectral resolution we shall establish will be denoted by $\{U_x\}$. Precisely speaking we have a Hilbert space (not necessarily separable) and for each x in G we have a unitary operator U_x acting on this Hilbert space, and $U_x U_y = U_{x+y}$ for all x, y . Points in this Hilbert space will be denoted by α, β and γ . We shall assume in proving our spectral theorems that U_x is weakly continuous in x , i.e., that $(U_x \alpha, \beta)$ is continuous in x for each α and β .

The term "operator" will always mean a bounded linear everywhere-defined transformation of this Hilbert space into itself, and if V is an operator then $||| V |||$ will denote its bound. If V is an operator such that $(V\alpha, \beta) = (\alpha, V\beta)$ for all α, β , then we shall call V real, or Hermitian. If V and V' are Hermitian operators with $(V\alpha, \alpha) \geq (V'\alpha, \alpha)$ for all α , then we shall write $V \geq V'$. We recall here the well-known fact that, if $L(\alpha, \beta)$ is a bounded Hermitian-bilinear functional, then there exists a Hermitian operator V with $L(\alpha, \beta) = (V\alpha, \beta)$.

Before going on to the technical details of the paper we make the following comments. Our proof of Theorem 1 uses the Herglotz-Bochner-Weil representation theorem in the same way as Bochner [1] previously used the corresponding theorem for functions of a real variable to obtain Stone's theorem. If U_x is group translation on G (i.e., our Hilbert space is $L_2(G)$, where G is the group over which the parameter x ranges and U_x is defined by $U_x f(y) = f(yx)$), then the Fourier transform V carries U_x into a group U'_x acting on $L_2(G^*)$. This group U'_x is that defined by $U'_x \varphi(\xi) = [x, \xi] \varphi(\xi)$ and its spectral resolution is that defined by $E(B^*) \varphi = \varphi_{B^*} \cdot \varphi$, where φ_{B^*} is the characteristic function of B^* . It is interesting to ask if any similarity with this situation remains in the case of non-Abelian groups since there one has both the group of unitary operators given by group translation and the spectral family defined on this same way. Only one has no related U'_x and no dual group. In a later paper we hope to show that these matters, in the case of compact groups, are related to the Tannaka duality theorem and also have some meaning for non-compact locally compact non-Abelian groups.

2. The spectral theorem. By a *spectral family* on G^* we shall mean a family of operators $\{E(B^*)\}$, one for each Borel set B^* , with the following properties:

- (1) Each $E(B^*)$ is a projection operator.
- (2) $E(\text{null set}) = 0$.
- (3) $E(G^*) = I$.
- (4) $E(B_1^*)E(B_2^*) = E(B_1^*B_2^*)$.
- (5) If $B^* = \sum B_i^*$ and the B_i^* are disjoint, then $E(B^*) = \sum E(B_i^*)$.
- (6) $E(B^*) = \inf E(O^*)$, this inf being taken over all open O^* containing B^* (and this equality meaning, too, that this inf always exists).

We can now state and prove our principal spectral theorem:

THEOREM 1. *If $\{U_x\}$ is weakly continuous, then there is a unique spectral family $\{E(B^*)\}$ such that*

$$(2.1) \quad U_x = \int [x, \xi] E(d\xi)$$

(this equality holding in the sense that the usual approximating sums for an integral, when formed in terms of the operators and functions under this integral sign, converge uniformly to U_x). An operator commutes with every U_x if and only if it commutes with every $E(B^*)$.

Proof. It is trivial that, for each α , $(U_x\alpha, \alpha)$ is a positive definite function and hence, by I, has an integral representation:

$$(2.2) \quad (U_x\alpha, \alpha) = \int [x, \xi] m_\alpha(d\xi).$$

Because $(U_x\alpha, \beta)$ is a linear combination of terms of the form $(U_x\gamma, \gamma)$, by taking $\gamma = \alpha, \beta, \alpha + \beta$, and $\alpha + i\beta$, it follows that for each α and β there exists a complex measure $m_{\alpha, \beta}$ such that

$$(2.3) \quad (U_x\alpha, \beta) = \int [x, \xi] m_{\alpha, \beta}(d\xi).$$

The uniqueness of this representation then implies that (for each Borel set B^*) $m_{\alpha, \beta}(B^*)$ is a bilinear functional and also

$$(2.4) \quad m_{\alpha, \beta}(B^*) = \overline{m_{\beta, \alpha}(B^*)}.$$

Such a functional satisfies the Schwarz inequality:

$$(2.5) \quad |m_{\alpha, \beta}(B^*)|^2 \leq m_{\alpha, \alpha}(B^*)m_{\beta, \beta}(B^*).$$

Remembering that $m_\alpha(B^*) = m_{\alpha, \alpha}(B^*)$ is a measure and using (2.2) with $x = 0$, we have

$$(2.6) \quad m_{\alpha, \alpha}(B^*) \leq m_\alpha(G^*) = \|\alpha\|^2.$$

Then (2.5) and (2.6) together show that $m_{\alpha, \beta}(B^*)$ is a bounded, as well as bilinear, functional. We conclude that for each Borel set B^* there is a real operator (it is real because of (2.4)) $E(B^*)$ such that $(E(B^*)\alpha, \beta) = m_{\alpha, \beta}(B^*)$.

We shall now show that, if B_1^* and B_2^* are any two Borel sets, then $E(B_1^* B_2^*) = E(B_1^*)E(B_2^*)$. We have

$$(U_{x+y}\alpha, \beta) = \int [x, \xi][y, \xi](E(d\xi)\alpha, \beta)$$

so defining the complex measure $n(B^*)$ by

$$(2.7) \quad n(B^*) = \int_{B^*} [y, \xi](E(d\xi)\alpha, \beta),$$

we have

$$(U_{x+y}\alpha, \beta) = \int [x, \xi] n(d\xi).$$

We also have

$$(2.8) \quad (U_{x+y}\alpha, \beta) = \int [x, \xi](E(d\xi)U_y\alpha, \beta).$$

By the uniqueness theorem we conclude that, for every Borel set B_1^* ,

$$(2.9) \quad \int_{B_1^*} [y, \xi](E(d\xi)\alpha, \beta) = (E(B_1^*)U_y\alpha, \beta).$$

This is the same as saying that, if we define $F(B^*)$ (for all Borel sets B^* in terms of an arbitrary fixed Borel set B_1^*) by

$$(2.10) \quad F(B^*) = E(B^* B_1^*),$$

then

$$(2.11) \quad \int [y, \xi](F(d\xi)\alpha, \beta) = (E(B_1^*)U_y\alpha, \beta).$$

We also have, however,

$$(2.12) \quad (E(B_1^*)U_y\alpha, \beta) = (U_y\alpha, E(B_1^*)\beta) = \int [x, \xi](E(d\xi)\alpha, E(B_1^*)\beta).$$

From (2.11) and (2.12) we conclude that (for any Borel set B^*)

$$(2.13) \quad (F(B^*)\alpha, \beta) = (E(B^*)\alpha, E(B_1^*)\beta).$$

Using (2.10), (2.13) and the reality of the $E(B_1^*)$, we conclude that $E(B_1^* B_2^*) = E(B_1^*)E(B_2^*)$.

The last paragraph showed that our family $\{E(B^*)\}$ has properties (1) and (4) of a spectral family, and from the fact that our original $m_\alpha(B^*)$ was a regular measure we conclude that it has properties (2), (5) and (6). Property (3) is implied by (2.2). Hence our family $\{E(B^*)\}$ is a spectral family.

We already have (2.2) and it is easy to show that the integral

$$\int [x, \xi] E(d\xi)$$

exists (in the sense mentioned in the statement of this theorem) so we conclude that (2.1) is true. The uniqueness of the spectral family $\{E(B^*)\}$ is clear because if a spectral family satisfies (2.1) then it will satisfy (2.2) and we already know the uniqueness of the representation (2.2).

Now let V be any operator. If V commutes with every U_x , then $(U_x V\alpha, \beta) = (U_x \alpha, V^* \beta)$ and hence by the uniqueness of our representation we have $(E(B^*) V\alpha, \beta) = (E(B^*) \alpha, V^* \beta)$, so V commutes with every $E(B^*)$. If V commutes with every $E(B^*)$ we can show in the same way that V commutes with every U_x . Hence the theorem is proved.

3. The operational calculus. We shall say that a Borel function $\varphi(\xi)$ is essentially bounded (with respect to the spectral family $\{E(B^*)\}$ of the given group $\{U_x\}$) if there exists a constant k such that $E(|\varphi| > k) = 0$. For each such φ we define an operator V_φ by

$$(3.1) \quad V_\varphi = \int \varphi(\xi) E(d\xi)$$

(where again the integral stands for the limit, in the sense of the strong topology of operator space, of the usual approximating sums).

THEOREM 2. *The correspondence $\varphi \rightarrow V_\varphi$, defined for all essentially bounded Borel functions by (3.1) has the following properties:*

- (a) $V_{a\varphi+b\psi} = aV_\varphi + bV_\psi$,
- (b) $V_{\varphi\psi} = V_\varphi V_\psi$,
- (c) $V_{\bar{\varphi}} = V_\varphi^*$,
- (d) if φ is real, then V_φ is real,
- (e) if $\varphi \geq \psi$, then $V_\varphi \geq V_\psi$,
- (f) if $|\varphi| \equiv 1$, then V_φ is unitary,
- (g) if φ = the characteristic function of the Borel set B^* , then $V_\varphi = E(B^*)$,
- (h) $\|V_\varphi\| \leq$ the essential bound of $|\varphi|$,
- (i) if $\varphi(\xi)$ is the Fourier transform of $f(x) \in L_1$, then (for all α and β),

$$(V_\varphi \alpha, \beta) = \int f(x) (U_x \alpha, \beta) dx.$$

The proof is obvious.

4. Spectral resolution for subgroups and quotient groups. In this section we shall be concerned with the spectral resolution of a group of unitary operators $\{U_x\}$ in case the group parameter x runs over some group, but U_x really depends

on some subgroup or quotient group of the full group. We set up the following notation:

$\bar{G}$ = group, with points $\bar{x}$,
 G = closed subgroup of $\bar{G}$, with points x ,
 $\bar{G}^*$ = character group of $\bar{G}$, with subsets $\bar{B}^*$ and points $\bar{\xi}$,
 G' = that subgroup of $\bar{G}$ which annihilates G ,
 G^* = $\bar{G}/G'$ = character group of G , with subsets B^* and points ξ ,
 $\{U_{\bar{x}}\}$ = group of unitary operators, defined for $\bar{x} \in \bar{G}$,
 $\{\bar{E}(\bar{B}^*)\}$ = spectral resolution of $\{U_{\bar{x}}\}$.

As before, we assume that $\bar{G}$, and hence all the other groups here, are locally compact and Abelian.

We shall have to consider the natural mapping of $\bar{G}^*$ into G^* , taking each $\bar{\xi} \in \bar{G}^*$ into that coset of $\bar{G}^*/G'$ to which $\bar{\xi}$ belongs. We define, for each Borel set B^* , the operator $E(B^*)$ by

$$(4.1) \quad E(B^*) = \bar{E}(\bar{B}^*),$$

where $\bar{B}^*$ is the origin of B^* under the mapping $\bar{G}^* \rightarrow G^*$.

When we are given $\{U_{\bar{x}}\}$, $\bar{x}$ running through $\bar{G}$, we can consider the subgroup $\{U_x\}$, x running through G . The following theorem gives the spectral resolution of $\{U_x\}$ in terms of the spectral resolution of $\{U_{\bar{x}}\}$.

THEOREM 3. *The spectral resolution of $\{U_x\}$ is the family $\{E(B^*)\}$ defined by (4.1).*

Proof. It can be immediately verified from (4.1) that $\{E(B^*)\}$ has the properties (1)-(6) of §2, and hence is a spectral family. We have

$$U_{\bar{x}} = \int [\bar{x}, \bar{\xi}] \bar{E}(d\bar{\xi}).$$

For $\bar{x} = x \in G$, the function $[x, \bar{\xi}]$ is a function of G^* , i.e., it depends only on the cosets $G' + \bar{\xi}$, so by (4.1) we can write

$$U_x = \int [x, \xi] E(d\xi).$$

The uniqueness of the spectral resolution then shows that $\{E(B^*)\}$ is the spectral resolution of $\{U_x\}$.

We now consider $\{U_{\bar{x}}\}$ for all $\bar{x} \in \bar{G}$ but suppose that $U_{\bar{x}}$ depends only on $\bar{G}/G$, i.e., U_x is constant on the cosets $G + x$. The following theorem shows that in this case the spectrum is contained in G' .

THEOREM 4. *If $U_{\bar{x}}$ depends only on $\bar{G}/G$, then, for any Borel set B^* , $E(B^*) = E(B^*G')$.*

Proof. We consider the subgroup $\{U_x\}$ with $x \in G$. Since $U_x = I$ for $x \in G$ the spectral resolution of $\{U_x\}$ is defined by

$$(4.2) \quad E(B^*) = I \text{ if } 0 \in B^*, \quad E(B^*) = 0 \text{ if } 0 \notin B^*.$$

By Theorem 3 we have

$$(4.3) \quad E(B^*) = \overline{E}(\overline{B}^*),$$

where $\overline{B}^*$ is the origin of B^* under the natural mapping of $\overline{G}^*$ onto G^* . Since $0 \in B^*$ means $G' \subseteq B^*$ and $0 \notin B^*$ means $G'B^* = 0$, (4.2) and (4.3) together establish the theorem.

REFERENCES

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